Convention: power rails above components; GND below.

J1
V24_RAW
GND
Conn_01x02

F1
1A 5x20
V24_RAW
V24_AFTER_FUSE

D1
SS24
V24_AFTER_FUSE
V24_FUSED

TVS1
SMAJ50CA
V24_FUSED
GND

R3
100k
V24_FUSED
Q1_GATE

Q1
IRLML6344TRPBF
V24_SW
Q1_GATE

R4
100k
PWR_EN
GND

Q2
IRLML6344TRPBF
PWR_EN
GND

V24_FUSED

R5
1M
V24_FUSED
V24_SENSE

R6
110k
V24_SENSE
GND

C5
100nF
V24_SENSE
GND

U1
TPS62933FDRLR
V24_SW
U1_BST
U1_SW
V3V3_SW
GND

C1
22uF/25V
V24_SW
GND

C_BST
100nF
U1_BST
U1_SW

L1
2.2uH
U1_SW
V3V3_SW

C2
22uF/25V
V3V3_SW
GND

C3
22uF/35V
V24_SW
GND

U2
R-78E12-1.0
V24_SW
GND
V12_CAT5E

C4
22uF/25V
V12_CAT5E
GND

U3
SN65HVD3082E
V3V3_SW
UART_RX_3V3
DE_RE
UART_TX_3V3
GND
RS485_A
RS485_B

R11
680
V3V3_SW
RS485_A

R12
680
RS485_B
GND

R10
120
RS485_A
RS485_B

TVS2
SMAJ12CA
RS485_A
RS485_B

R7
10k
V3V3_SW
ESP_EN

C8
1uF
ESP_EN
GND

C6
10uF
V3V3_SW
GND

C7
100nF
V3V3_SW
GND

MOD1
ESP32-S3-WROOM-1-N16R8
V3V3_SW
ESP_EN
V24_SENSE
DE_RE
PWR_EN
I2C_SDA
I2C_SCL
BTN_OVERRIDE
DBG_UART_TX
DBG_UART_RX
UART_TX_3V3
UART_RX_3V3
USB_DM
USB_DP
PSRAM
GND

J2
RJ45
GND
RS485_B
RS485_A
V12_CAT5E

J3
USB-OTG
USB_DP
USB_DM
ESP_EN
GND

J5
JART-DBG
DBG_UART_TX
DBG_UART_RX
GND
ESP_EN

R9
4.7k
V3V3_SW
I2C_SCL

R8
4.7k
V3V3_SW
I2C_SDA

BAT1
CR2032
V_BAT_RTC
GND

RTC1
DS3231SN#
V3V3_SW
V_BAT_RTC
I2C_SCL
I2C_SDA
GND

C9
100nF
V3V3_SW
GND

BTN1
OVERRIDE
BTN_OVERRIDE
GND

R13
1M
V3V3_SW
BTN_OVERRIDE

C11
100nF
BTN_OVERRIDE
GND

#FLG1
PWR_FLAG
V24_FUSED

#FLG1
PWR_FLAG
GND

#FLG1
PWR_FLAG
V24_SW

#FLG1
PWR_FLAG
V3V3_SW

#FLG1
PWR_FLAG
V_BAT_RTC

Volthium

Sheet: /
File: battery_side.kicad_sch

Title: Volthium reader — battery side

Size: A3	Date: 2026-05-24	**Rev: CP-schematic-cleanup**
KiCad E.D.A. 10.0.3		Id: 1/1